

Owen Kim

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EDUCATION

University of Waterloo

Sep. 2024 – May 2029

BASc Biomedical Engineering – GPA: 3.91/4.0 (89% Cumulative Average)

Relevant Courses: Systems and Signals, Circuits, Data Structures and Algorithms, Statistics

EXPERIENCE

The Hospital for Sick Children

Jan. 2026 – Present

R&D Engineer

Toronto, ON

- Led a cross-lab initiative across **9+ researchers and engineers** spanning neuroscience and hardware teams to evaluate transcranial focused ultrasound neuromodulation and OPM-MEG neural monitoring.
- Identified and mitigated electromagnetic interference using shielded coaxial cabling and signal filtering.
- Drove prototyping of a wearable neuromodulation helmet in SolidWorks, iterating through **11+ transducer/sensor configurations** and validating device integration through phantom testing and simulation.
- Co-authored a **\$225,000** internal research grant and coordinated technical priorities across **3 research groups**.

Vision & Image Processing Lab and Critical ML Lab

Aug. 2025 – Present

Undergraduate Research Assistant

Waterloo, ON

- Implemented **active learning** pipelines in Python and PyTorch, applying **reinforcement learning-based** sampling strategies to a brain tumor imaging dataset for semantic segmentation.
- Co-authored two research manuscripts on active learning and preprocessing for medical image segmentation.

Harvard Medical School – Tearney Lab

May 2025 – Aug. 2025

Machine Learning Research Intern

Boston, MA

- Developed a **deep learning** pipeline for Parkinson's disease diagnosis through segmentation of esophageal OCT images.
- Translated clinician feedback into **16+ design requirements**, informing model architecture, training strategy, and evaluation criteria.
- Trained and validated a **LT-U-Net** model, achieving over **85% classification accuracy** across all classes.
- Presented research at the **Wellman Center for Photomedicine** at Massachusetts General Hospital.

PROJECTS

Myoelectric Prosthetic Arm | Signal Processing, Circuit Design, EMG, C++, KiCad

- Designed a **mixed-signal EMG acquisition and control system**, developing a 7-stage analog front-end for amplification, 20–450 Hz bandpass filtering, and digital triggering.
- Developed deterministic, non-blocking **C++ firmware** using a state machine to detect sustained muscle activation, control motor actuation, and implement timing-based trigger rejection.
- Designed a custom **KiCad PCB** integrating the analog front-end, microcontroller, and motor control.

United Mobility | Product Development, Software, Systems

- Co-founded a decentralized e-scooter platform, leading execution across **engineering, and product design**.
- Drove **cross-functional** development and deployment strategy across software and business operations.
- Awarded the **People's Choice Award** at the University of Waterloo Velocity Future Cities Pitch Competition.

TECHNICAL SKILLS

Programming: Python, C++, C#, Java, MATLAB, SQL

ML / Data: PyTorch, Deep Learning, Computer Vision, Semantic Segmentation, Active Learning

Signal Processing: MNE-Python, Digital Filter Design, Analog Signal Conditioning, Embedded DSP

Engineering: KiCad, SolidWorks, Git, Linux, PCB Design, Embedded Systems, Prototyping & Validation

Systems & AI Tools: Claude Code, Data Structures, Systems Integration, BoM Development, QFD

LEADERSHIP

BioTEC Conference

Oct. 2024 – Present

Logistics Director

Waterloo, ON

- Managed **20+ executives** and led **logistics** for Canada's largest student-run biotech conference, with over **250 attendees**.